Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/14/2022
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Can Polyamines Mitigate Plant Stress Response under Microgravity Conditions?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/16/2019	**End Date:**	09/15/2023
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/15/2023 per NSSC information (Ed., 7/26/22).		
Key Personnel Changes/Previous PI:	July 2020 report: Dr. Shih-Heng Su is an Assistant Scientist in the Masson lab who has been contributing to this project since its inception.		
COI Name (Institution):	Su, Shih-Heng Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC19K1483		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space exploration requires the design and implementation of bioregenerative life-support systems capable of maintaining breathable air, cleaning water, and producing food and fiber during space missions. Plants are expected to play key roles in these life-support systems. Therefore, it is important to understand the effects the microgravity environment has on plants to allow the design and development of better life-support systems for space exploration. Multiple studies have demonstrated that plants grown under microgravity display signs of stress, including altered morphology and expression profiles expected for stress response. Therefore, the development of mitigation strategies aimed at alleviating such stress will be important. Previous studies by multiple research teams including ours have demonstrated a role for polyamines in stress mitigation in plants. Polyamines are a group of compounds whose abundance increases in plants under stress. These signaling molecules have also been shown to contribute to stress mitigation in ground-based experiments. Therefore, we hypothesize that polyamines might also contribute to stress mitigation under microgravity, and propose to test this hypothesis by investigating the growth and expression profiles of Arabidopsis thaliana seedlings engineered to modify the pool of putrescine (one of these polyamines) accumulating in their tissues. Specific objectives will include: 1) Comparing the growth of Arabidopsis thaliana seedlings with genetic alterations affecting the pool of putrescine to that of wild type, within the VEGGIE hardware, both under microgravity on the International Space Station (ISS) and on the ground (1-g control), and 2) Use RNAseq approaches to investigate the effect of microgravity relative to 1-g ground control on the transcriptomes of wild type and mutant seedlings with alterations in their putrescine pool. These investigations should lead to a better understanding of the roles played by polyamines in stress mitigation under microgravity. It may identify novel genetic engineering methods that improve plant growth under microgravity, thereby allowing development of optimized bioregenerative life-support systems for long-term space exploration missions. This project may also have important applications in agriculture for crop production on marginal lands. Finally, our experiments will yield large sets of transcriptomic data from previously uncharacterized genotypes, which will be added to the GeneLab dataset.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project will lead to a better understanding of the role of Putrescine and/or its derived signaling products in mitigating the stress associated with growth under microgravity environments such as those encountered during spaceflight and on the International Space Station. It may also allow the identification of novel genetic engineering strategies aimed at improving plant adaptation to the microgravity environment, with potential impact on our ability to improve the genome of other plant species / crops for utilization in bioregenerative life-support systems during spaceflight. A better understanding of plant responses to stress may also have important applications in agriculture for crop production on marginal lands. New large-scale RNAseq expression data associated with growth under microgravity for previously untested Arabidopsis genotypes will also be generated during this project and submitted to the GeneLab Data System. This project will allow us to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, and also generate novel instructional materials allowing effective teaching of the difficult concepts of gravity perception by plants and their importance for agricultural production on Earth and during spaceflight. Such materials should be usable in K-12, undergraduate, and graduate education, as well as general-public outreach.
	SUMMARY The main objective of APEX-08 is to evaluate the impact of genetic alterations of the putrescine pool in Arabidopsis thaliana on the seedlings' ability to grow, develop, and respond transcriptionally to the microgravity conditions of the International Space Station (ISS) relative to ground control. To this end, we previously reported on the development and characterization of Arabidopsis thaliana mutant and transgenic lines with alterations in putrescine metabolism, including: (1) wild type Col-0, (2) adc1-1 mutant affected in its ability to synthesize Put; (3) Cuao3-1 and cuao3-100cdr7 mutants affected in their ability to catabolize putrescine; (4) transgenic plants over-expressing the Poncirus trifoliata ADC (PtADC) gene involved in putrescine synthesis (hereafter named PtADC-OX); and (5) transgenic cuao3 plants over-expressing the AtCuAO3 gene (hereafter named cuao3-1[CuAO3]-OX). These genotypes should allow evaluation of the effect of putrescine-pool alterations on Arabidopsis thaliana responses to the stress encountered under microgravity. In 2021, we reported on experiments that allowed us to bulk up seed material from these wild type, mutant, and over-expression lines for verification tests as well as for the flight and ground control experiments, and successfully completed both the Science Verification (SVT) and Experimental Verification (EVT) tests at NASA Kennedy Space Center (KSC). Preparation for these tests and their implementation allowed us to optimize the experimental approach and success criteria for APEX-08. Consequently, we were able to implement both the APEX-08 flight experiment on the International Space Station (ISS) and its ground control at KSC in the fall of 2021. As discussed below, both experiments went smoothly, and their results are currently being analyzed. PROGRESS MADE DURING THE 2021-22 PERIOD 1) Implementation of APEX-08 1.a. Experimental Approach: The APEX-08 experiment was carried out following the approach we optimized during the SVT (November 2020) and EVT (March 2021). Briefly, 10x10" square Petri dishes were poured with 1/2MS medium containing 0.8% agar at KSC on 8/24/2021. The next day, seeds from each genotype were surface-sterilized and plated on the surface of the medium, in a line, at a density of 50 seeds per plate. Each genotype was represented by 5 plates (5 replicates). Seeded plates were immediately treated with far-red light in the cold room in darkness, then wrapped in aluminum foil and stored at 4°C until handed over to the Multi-Element Integrated Test (MEIT) team for cold storage the next day. Upon transfer into the Dragon CRS-23 capsule, the experiment took off on SpaceX CRS-23 (SpX-23) on 8/29/2021. After reaching the ISS, the plates were unwrapped and inserted into the NASA Vegetable Production System (Veggie) growth unit by astronaut Shane Kimbrough on 9/1/2021, following the blueprint previously established for the SVT. Pictures of each plate were taken before insertion, to make sure seed germination did not occur during takeoff and the plates were not cracked. Germination was then triggered by a 24-h red light treatment. After one day, both green and blue lights were turned on (low setting) and the seedlings were allowed to germinate and grow for an additional 8 days. The temperature, humidity, and CO2 profiles were recorded with HOBO sensors located near the plates on Veggie.

Task Progress:

At the end of this growth period, plates were photographed, and the seedlings were harvested into RNAlater within NASA Kennedy Fixation Tubes (KFTs). After 1 day at room temperature, the KFTs were transferred into the -80°C freezer until returned to the ground on 9/30/21.

A ground control (GC) mimicking all aspects of the flight experiment was carried out at KSC, with a 48-h delay. Here again, the 30 plates submitted to this experiment were photographed both at the time of insertion and after the growth period. Seedlings were harvested in RNAlater within KFTs at the end of the experiment, then frozen at -80°C for storage and transported to the Principal Investigator (PI) laboratory for further analysis.

At the end of the experiment, the photographs were used to evaluate the effect of microgravity on seedling growth and morphology relative to ground control, and harvested tissues were used to extract RNA for expression profiling by RNAseq. Preliminary results from these experiments are summarized below.

1.b. Preliminary Results:

1.b.1. Far-red light pretreatment inhibits the germination of plated seeds

To verify the effectiveness of far-red light pretreatment at inhibiting the germination of plated seeds and also ensure that seeds remained attached to the medium and the plates did not crack during takeoff, we analyzed the photographs taken by astronaut Shane Kimbrough at the time of insertion on the ISS, and by the MEIT team for the ground control at KSC. In both cases, there was no evidence of cracked plates, detached or germinated seeds, indicating the experiment could proceed as planned.

1.b.2. A 9-day growth period in Veggie on the ISS (Flight experiment, FL) and on the ground at KSC (ground control, GC) led to high germination rates and production of significant amounts of plant biomass, adequate for initial morphometric analysis as well as RNA extraction and RNAseq analysis.

1.b.2.a. Morphometric Analysis

Analysis of the photographs taken at the end of the growth period for both flight and ground-control groups indicated close to 100% germination for all genotypes tested, although it was not possible to assign a precise germination rate because the seedlings were intertwined at this late stage of growth due to the high-density plating necessary to obtain sufficient material for RNA extraction and subsequent analysis.

In both the flight and ground-control experiments, the seedlings grew well, eventually covering most of the plate surfaces. A very noticeable difference between flight and ground control seedlings of all genotypes was a general tendency for the roots to grow downward in the ground control, and more randomly in the flight experiment.

The high density of seedlings within each plate prevented us from performing a morphometric analysis of organs' growth using the photographs taken at the end of the growth period. However, upon receipt of the fixed and frozen tissue samples from KSC at the end of the experiment, we were able to retrieve a few seedlings from the bulk before RNA extraction, which we flattened on a glass surface and used to evaluate shoot sizes. This analysis was very informative, indicating a significant increase in green-leaf area and petiole sizes for both cotyledons and the first two true leaves in microgravity-exposed samples relative to ground controls. The differences in petiole sizes were significant for all genotypes tested. The cotyledon petiole size ratio between flight and ground control samples, on the other hand, showed no significant differences between genotypes. The latter observation may be explained by the fact that cotyledons are embryonic organs that formed during seed development, before microgravity exposure. Finally, looking at total leaf area, adc1-1 mutant seedlings displayed smaller values under ground-control conditions, but this difference was completely alleviated under flight conditions, suggesting that a decrease in arginine decarboxylase activity in the mutant (likely to result in decreased putrescine synthesis) did not affect its ability to develop biomass relative to wild type under microgravity conditions.

To identify possible border-effects on measured morphological phenotypes, we used ANOVA. Results indicate the existence of only minor effects from the corner positions within Veggie in the flight experiment, which were not recapitulated on the ground.

1.b.2.b. RNA extraction and RNAseq analysis

Half of the fixed tissue materials returned to the lab after harvesting in both the flight experiment on the ISS and the ground control at KSC, were recovered, dissected for separation of shoot and root tissues, and subjected to RNA extraction. Extracted RNAs were then analyzed using a combination of Nanodrop and TapeStation procedures. Overall, RNAs of sufficient quality for library construction and RNAseq analysis were obtained from all spaceflight and ground-control tissues.

After rRNA reduction, these RNA samples were subjected to TruSeq library construction, and sequencing by the Wisconsin Gene Expression Center at the University of Wisconsin Biotechnology Center. Sequence reads are currently being analyzed.

2) Conclusion

The experimental conditions that were optimized during our Science- and Experiment-Verification Tests at KSC last year allowed successful implementation of both our flight-ISS and ground-control experiments. In both cases, all six genotypes displayed excellent germination and growth, meeting the experimental criteria defined in the NASA Environmental Resources Document (ERD). The seedlings of cuao3-100cdr7 grew more slowly than those of the five other lines, as expected from our initial results during the SVT and EVT as well as other experiments in the lab.

The high plant density chosen for APEX-08 prevents careful biometric analysis of plant morphology and development. However, we were able to evaluate the effect of spaceflight on the morphology of the shoots by recovering several seedlings from the fixed and frozen material returned to the lab for RNA extraction and expression analysis, and we analyzed them for total leaf area, cotyledon, and first-two-leaves petiole sizes. Interestingly, all genotypes showed evidence of increased growth under microgravity relative to ground control: The total leaf area was larger in flight-exposed seedlings, in large part because the petioles were longer. For cotyledon petiole size, the ratio between flight-exposed and ground-control plants was similar for all genotypes tested, indicating exposure to the microgravity environment of the ISS had little impact on the development of these embryonic organs. On the other hand, a similar ratio for first-two-leaves petiole length was higher in adc1-1 and Col[CuAO3] over-expressing plants, suggesting a

	decrease in putrescine levels might facilitate leaf-petiole growth under microgravity. This conclusion seems compatible with the lower response ratio observed in the Col[PtADC1] over-expressing and cdr7 mutant lines. The RNAlater-fixed and frozen plant materials that had been harvested in both the flight experiment and the ground control were adequate for RNA extraction and RNAseq analysis, meeting the criteria established by the EVT ERD. They were subjected to RNAseq analysis at the Wisconsin Gene Expression Center. Sequence reads are currently being analyzed. They should provide exciting new clues to better understand the impact of both microgravity and alterations in the putrescine metabolic pathway on Arabidopsis seedling growth and stress responses.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Su S-H, Moen AD, Groskop RM, Baldwin K, and Masson PH. "Using low-speed 2-dimensional clinorotation to study gravi-sensitivity in Arabidopsis and Brachypodium." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021
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